

	Training (n=300)	Validating (n=128)	P-value
OS			
Mean (SD)	880 (752)	915 (864)	0.689
Median [Min, Max]	676 [14.0, 4500]	686 [6.00, 4230]	
age			
Mean (SD)	66.1 (13.3)	68.1 (12.3)	0.12
Median [Min, Max]	68.0 [31.0, 90.0]	71.0 [36.0, 90.0]	
status			
0	234 (78.0%)	100 (78.1%)	1
1	66 (22.0%)	28 (21.9%)	
gender			
male	156 (52.0%)	72 (56.2%)	0.449
female	144 (48.0%)	56 (43.8%)	
preoperative.CEA			
Mean (SD)	33.6 (197)	50.4 (218)	0.555
Median [Min, Max]	3.20 [0.200, 2610]	2.85 [0.00, 1390]	
Missing	106 (35.3%)	50 (39.1%)	
stage			
i	51 (17.0%)	22 (17.2%)	0.988
ii	117 (39.0%)	51 (39.8%)	
iii	90 (30.0%)	36 (28.1%)	
iv	42 (14.0%)	19 (14.8%)	
lymphatic.invasion			
no	167 (55.7%)	69 (53.9%)	0.909
yes	106 (35.3%)	45 (35.2%)	
Missing	27 (9.0%)	14 (10.9%)	
perineural.invasion			
no	98 (32.7%)	31 (24.2%)	0.513
yes	35 (11.7%)	8 (6.2%)	
Missing	167 (55.7%)	89 (69.5%)	
venous.invasion			
no	201 (67.0%)	82 (64.1%)	0.685
yes	61 (20.3%)	28 (21.9%)	
Missing	38 (12.7%)	18 (14.1%)	

Note:Chi-sq test for categorical variables,continuity correction is only used in the 2-by-2 case, and t-test for continous variables.